

Amirali D Shiraz

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Summary

Data engineer and machine learning practitioner with experience in Python, SQL, and large-scale data processing. Skilled in building ETL pipelines, feature engineering, and developing ML models for time-series and system-level data. Background in electrical engineering with hands-on experience in power systems and telemetry data. Fluent in English and French.

Work Experience

Bright Bee Technology Inc.

Data Scientist

03/2024 – Present

Toronto, Ontario, Canada

- Built end-to-end data pipelines in Python (Pandas, NumPy) and SQL to ingest, clean, and transform structured and unstructured electrical engineering data, converting raw technical content into analytics-ready datasets.
- Designed scalable ETL/ELT workflows using PySpark, Databricks, and AWS (S3, Glue), improving processing efficiency and reducing manual data preparation by 30%.
- Developed optimized data models and complex queries in SQL, working with large datasets to create reliable, high-performance tables for analytics and reporting layers (e.g., Tableau / Power BI).
- Worked closely with data scientists to perform feature engineering, data validation, and preprocessing, using scikit-learn and Python-based workflows to improve model performance and consistency.
- Designed and supported ML models on healthcare data (e.g., blood pressure, glucose levels) using scikit-learn / TensorFlow, enabling anomaly detection and fatigue prediction, contributing to early identification of potential risk conditions.

Polytechnique Montréal

Research Assistant

Jan 2021 - Sep 2024

Montreal, Quebec, Canada

- Processed and modeled data from over 140,000 electric water heater sensors, transforming raw electrical telemetry into clean, ML-ready time-series datasets using Python and SQL.
- Developed data-driven droop control strategies for grid frequency stabilization, combining MATLAB/Simulink simulations with Python-based statistical and ML methods.
- Applied stochastic modeling (Kolmogorov PDEs) to simulate large-scale grid behavior under uncertainty, improving system-level analysis and robustness.
- Integrated power monitoring devices with telemetry systems aligned with SCADA / EPMS environments, enabling reliable data collection and validation.

Rahbord Hooshmand

Intern

Jan 2018 – Dec 2020

- Performed statistical analysis on electrical and operational data using regression and Bayesian methods to support system performance improvements.
- Developed C++ components for telemetry systems in smart grid applications, improving modularity and system reliability.
- Optimized SQL queries and ETL pipelines for large-scale grid data, reducing processing latency by approximately 20%.
- Built dashboards using Power BI and Tableau to visualize system performance and support data-driven decision-making.

Education

Polytechnique Montréal

Master of Applied Science in Electrical Engineering, Montreal, QC, Canada

Sep 2021 – Sep 2024

University of Tehran

Bachelor of Science in Electrical Engineering, Tehran, Iran

Sep 2016 – Jun 2021

Projects

DMFlix – Dagster + Snowflake + ML Embeddings

Automated an ML pipeline with **Dagster** and **Snowflake** to decode embeddings, run t-SNE, and generate artifacts for

large-scale movie datasets. Built configurable assets with robust decoding, data cleaning, and feature augmentation, while hardening pipelines with parameterized SQL, resource-injected Snowflake connections.

Temporal Difference Learning for Real-Time Control

Implemented **SARSA** and **Q-learning** algorithms for stochastic control in unknown environments; trained RL agents with dynamic reward structures and benchmarked performance using Python and NumPy.

Multi-GPU Stable Diffusion Pipeline for Visual Ad Generation

Built and synchronized a custom GPU cluster for **Stable Diffusion 1.5** and **Stable Diffusion XL**, enabling parallelized image generation for advertising. Handled full system setup—GPU rack assembly, kernel tuning, NCCL sync—and optimized **CUDA pipelines** for batch performance. Engineered prompts and features to control output quality and style across multiple model checkpoints.

Real-Time Disaster Sentiment Analysis on AWS

Deployed a real-time **NLP pipeline** using **AWS Kinesis**, **Lambda**, and **Comprehend** to monitor social media sentiment during disasters; built a **QuickSight** dashboard to support live crisis response.

Bayesian and Distributed Multi-Agent Estimation

Developed a decentralized estimation system using **Bayesian modeling** and **distributed Kalman filters**; implemented real-time prediction across agents using Python, optimizing accuracy and communication in uncertain environments.

Git System Development in C

Designed a custom version control system in **C** with core Git functionalities (commit, log, checkout); built efficient data structures for snapshotting and delta comparison across file states.

Super Mario Game Recreation in C++

Rebuilt core mechanics of **Super Mario** in **C++** using object-oriented principles; implemented character physics, level progression, and collision detection to simulate 2D gameplay in a console environment.

Technical Skills

Programming & Data: Python (Pandas, NumPy), SQL, PySpark, Time-Series Data, Data Structures & Algorithms

Data Engineering & Cloud: ETL/ELT Pipelines, Databricks, AWS (S3, Glue, Kinesis, SageMaker), Data Modeling, Large-Scale Data Processing

Machine Learning & AI: Scikit-learn, TensorFlow, PyTorch, Time-Series Forecasting, Anomaly Detection, Reinforcement Learning

Electrical Systems: Power Systems, Grid Frequency Control, Droop Control, Demand Response, Sensors & Telemetry, SCADA/EPMS

Modeling & Visualization: MATLAB, Simulink, Statistical Modeling, Power BI, Matplotlib, Feature Engineering

Version Control & Workflow: Git, GitHub, CI/CD

Certifications

Data Analytics Specialization

Google, Nov 2024

Covered exploratory data analysis, dashboards, SQL queries, and predictive modeling using Google tools and open-source libraries.

Neural Networks and Deep Learning

DeepLearning.AI, Nov 2024

Completed practical training on forward/backward propagation, activation functions, gradient descent, and tuning multi-layer neural networks.

Machine Learning Pipelines With Azure ML Studio

Coursera Project Network, Sep 2023

Designed ML pipelines using Azure's visual tools, including preprocessing, model selection, deployment, and performance tracking.

Awards & Scholarships

Ranked Top 0.17% in National Entrance Exam (Iran)

Placed 250th out of over 150,000 participants nationwide, gaining admission to Iran's top-ranked engineering programs.

Full Graduate Funding – Polytechnique Montréal

Awarded full research funding to pursue M.A.Sc. at **Polytechnique Montréal**, ranked among the top 4 Electrical Engineering schools in Canada.

Project Day Award – University of Tehran

Recognized for technical innovation in undergraduate final project on simulation and modeling of residential energy consumption.